



Robust End-to-End Image Transmission with Residual Learning

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Introduction & Motivation

01

Shift to Deep Learning in Image Transmission

Deep learning (DL)-based image transmission at the physical layer has demonstrated significant improvements in performance compared to conventional digital transmission methods, enabling more efficient and adaptive communication.

04

6G and Semantic Communication

6G communication aims to meet extreme requirements by leveraging semantic communication, which focuses on transmitting the meaning of data rather than raw bits, drastically reducing data load and enhancing transmission efficiency.

02

Challenges with Physical Layer Solutions

Implementing physical layer solutions requires fundamental changes in established communication standards, such as those used in cellular networks, posing major practical and compatibility challenges.

05

Limitations of Existing Residual Coding Neural Networks

Current neural network-based residual coding methods often rely on traditional compression tools like BPG or JPEG, or transmit residuals without accounting for channel-induced errors, limiting their robustness in real-world noisy channels.

03

Application Layer as a Feasible Alternative

Application layer solutions provide a standards-compliant approach that avoids extensive changes to physical layer protocols, enabling easier integration and deployment within existing network infrastructures.

06

Our Proposed DS²C² Solution

We introduce **Deep Semantics Source Channel Coding (DS²C²)**, an end-to-end neural network solution for layered image transmission at the application layer, designed to be robust against end-to-end channel errors and improve residual image encoding through latent representation alignment.

Key Contributions of DS²C²

DL-based Semantic Transmission

DS²C² is a novel deep learning-driven multi-layer image transmission scheme operating entirely at the application layer, ensuring compatibility with existing standards for next-generation networks.

Robust Latent Representation

The scheme maps residual images into latent representations that structurally align with channel characteristics, enhancing robustness against end-to-end channel errors.

Disjoint Modular Architecture

DS²C² employs a disjoint architecture combining **Recursive GAN (RGAN)** and **Block Residual Network (BResNet)**, with SumNet at the receiver, enabling separate training and modularity for adaptability.

Demonstrated Effectiveness

Numerical and visual results confirm DS²C²'s effectiveness under diverse channel error conditions, showing significant preservation of image quality compared to benchmarks.

Complete NN Residual Encoding

Unlike prior methods, DS²C² uses neural networks exclusively for encoding the residual image via BResNet, integrating two consecutive GANs (RGAN) for superior synthesis and generation.

Enhanced Summation Accuracy

SumNet improves reconstruction accuracy by refining the summation of base and residual layers at the receiver, vital for error-prone channel conditions.

DS²C² Overall Architecture

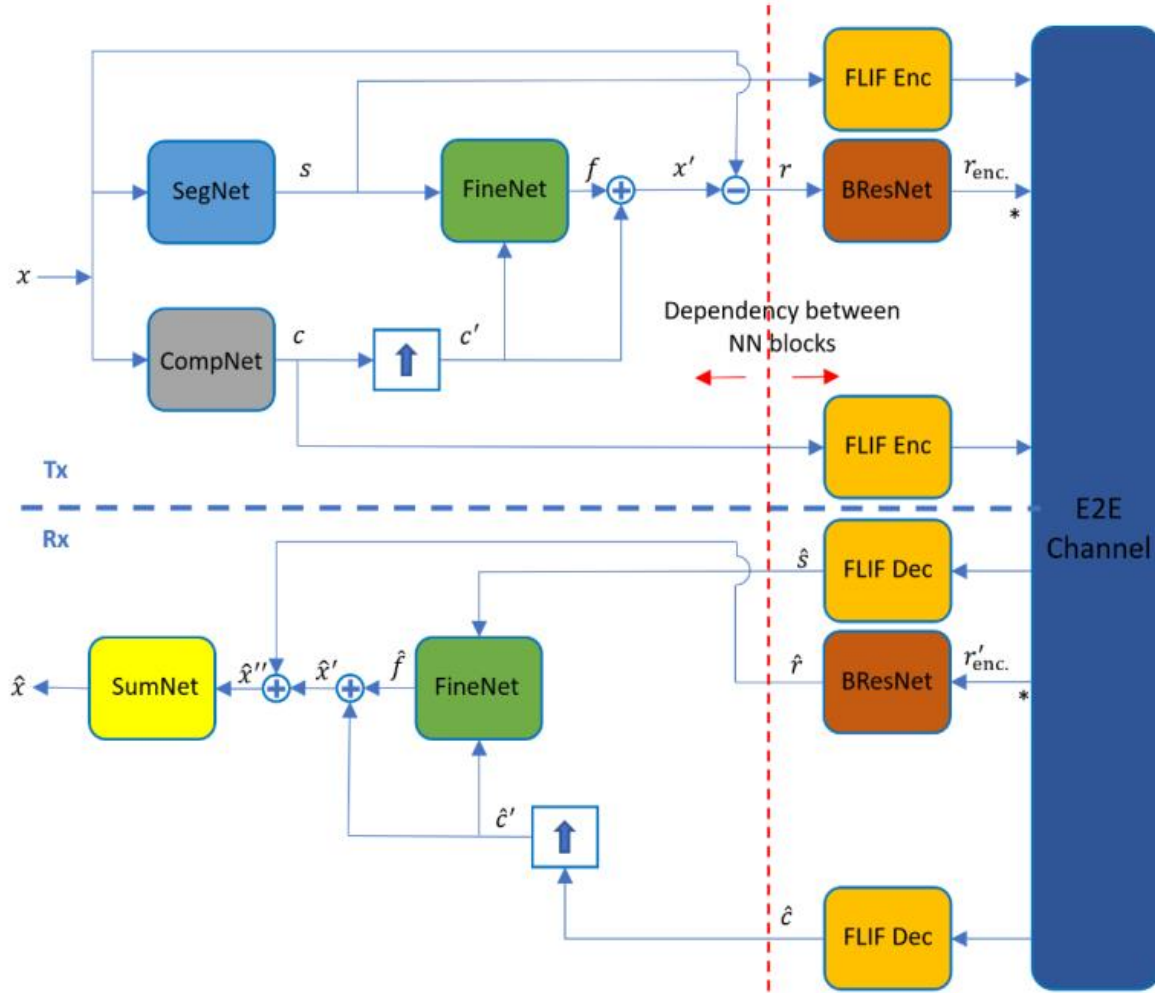


Fig. 1a DS²C² architecture

Base Layer and FineNet Synthesis

SegNet extracts semantic images (s) while CompNet downscales the image (c) which is upsampled to (c'). FineNet combines s and c' to produce (f), which is summed with (c') to generate the synthesized image (x')

Residual Encoding and Error Resilience

The residual image (r) is computed by subtracting (x') from the original image and encoded using BResNet. SumNet at the receiver refines summation accuracy.

RGAN Architecture, Quantization and Interleaving

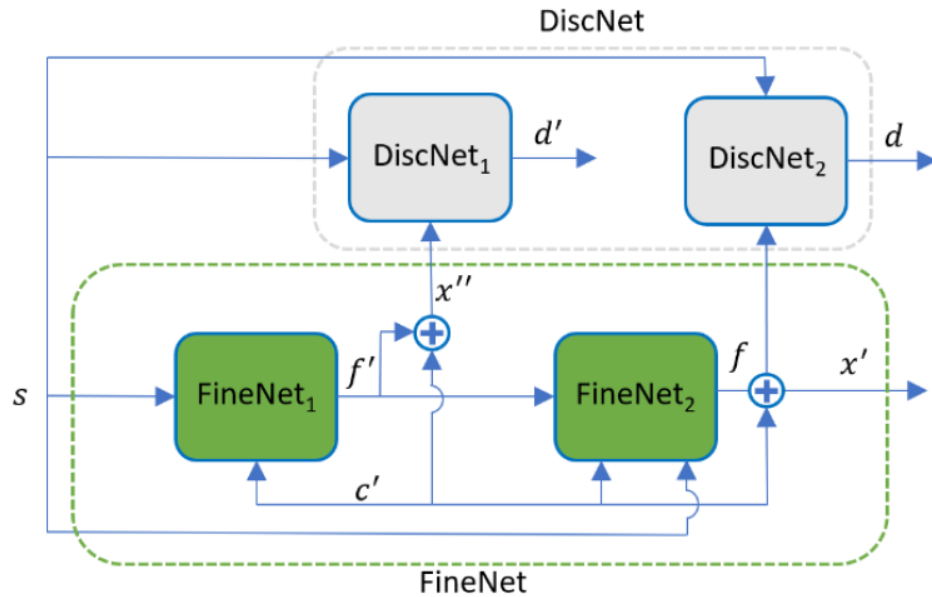


Fig. 1b RGAN architecture.

RGAN Components Overview

FineNet_i operates as the generator creating synthesized images, while DiscNet_i serves as the discriminator during training to improve realism. DiscNet_i is excluded during the testing phase.

Increasing the number of RGAN iterations (e.g., increasing from two to three consecutive GANs) improves image quality but at higher computational cost.

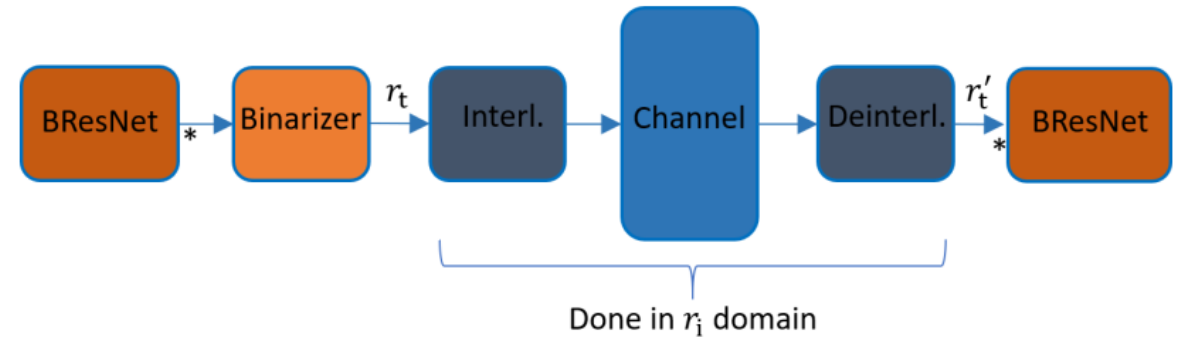


Fig. 2 Binarization, interleaving, channel error modelling, and deinterleaving operations used in DS²C².

Quantization and Interleaving

Quantization and pixel-level interleaving (to mitigate block channel errors) are applied.

Quantization, Interleaving & Channel Model

Quantization and Entropy Coding

Quantization and entropy coding techniques optimize the trade-off between compression rate and distortion, improving overall transmission efficiency.

Statistical Binarizer & Fixed BPP

A statistical binarizer is used to ensure a fixed bits-per-pixel (BPP) rate for consistent performance evaluation and controlled bitrate.

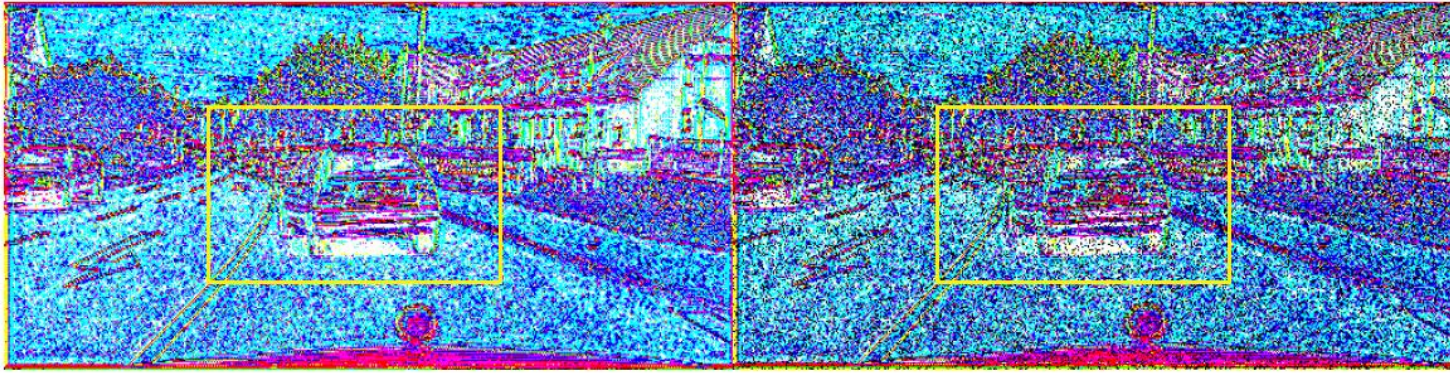
Block Binary Erasure Channel (BBEC)

The BBEC model simulates channel errors by erasing entire blocks of data, reflecting realistic burst error scenarios in wireless communication.

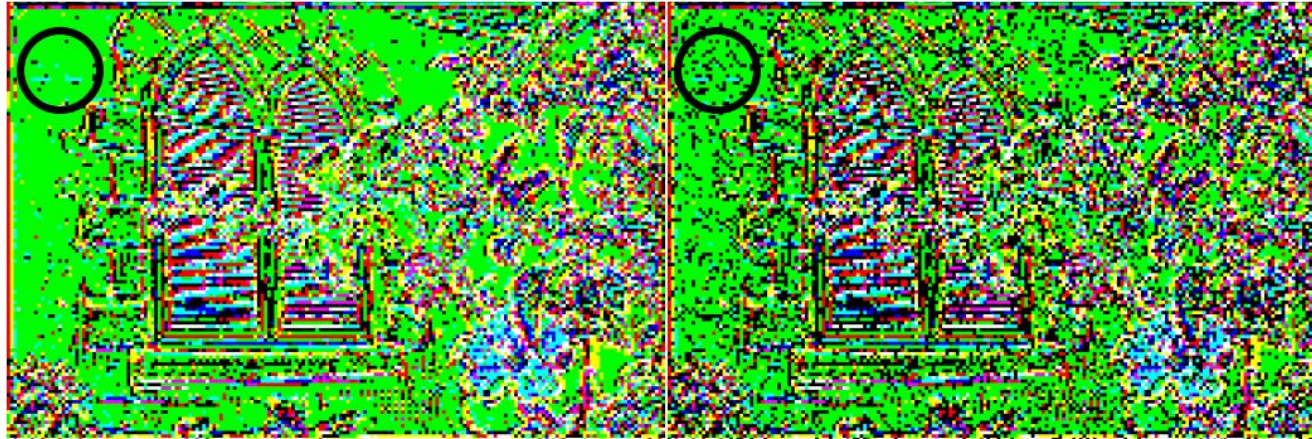
Pixel-Level Interleaver

Pixel-level interleaving redistributes data pixels across blocks to mitigate the impact of block losses, enhancing robustness against burst errors.

Robustness Mechanism of DS²C²



(a) Cityscapes dataset.



(b) Kodak dataset.

The residual image is encoded into a latent space designed to mimic the statistical noise patterns of the channel, enabling error resilience during transmission over error-prone networks.

Figure 3 shows similar noisy structures in the latent representation both before and after channel errors, demonstrating how DS²C² maintains signal integrity despite channel impairments.

Fig. 3 Similar structures observed in the latent representations of r_i (left) and r_i' (right) for Cityscapes and Kodak datasets.

Numerical Results: Datasets and Benchmarking

PSNR Comparison of DS²C² vs. LIC-GMM-AM Under Channel Errors

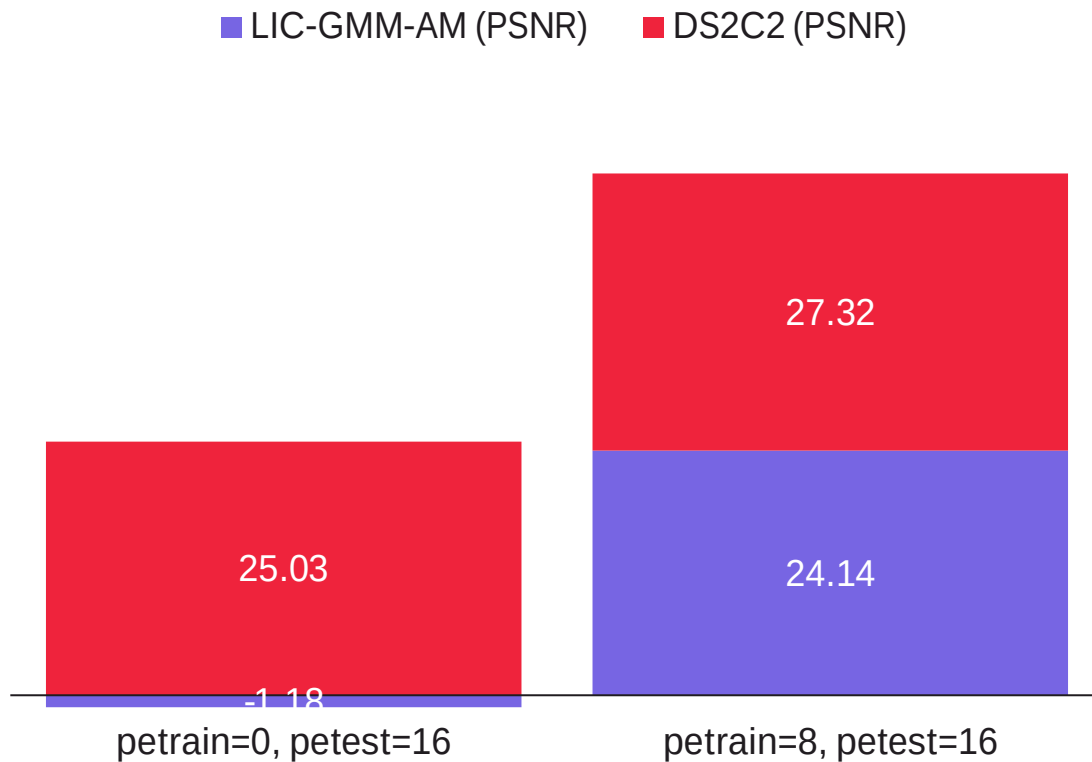


TABLE I: DS²C² vs. LIC-GMM-AM for Kodak dataset at 0.26 BPP.

Datasets and Benchmarking Highlights

- Datasets used include Cityscapes, ADE20K, and Kodak, providing diverse and challenging image scenarios for evaluation.
- LIC-GMM-AM shows significant PSNR degradation without channel error training (petrain=0, petest=16) dropping to -1.18 from 30.61.
- DS²C² achieves higher PSNR values of 25.03 (petrain=0, petest=16) and 27.32 (petrain=8, petest=16), outperforming LIC-GMM-AM in error-prone conditions.
- DS²C² reduces image quality degradation from about 20% to just 3%, demonstrating enhanced error resilience and effective residual coding.

Training Efficiency: Low Number of Training Images



(a) Only vehicles.



(b) Vehicles and pedestrians.

Fig. 4 Estimated images of DS²C² for Cityscapes dataset at 0.54 BPP for varying number of training images. Left: 16. Right: 2968.

DS²C² demonstrates remarkable capability to be quickly updated using a small number of training images, which is essential for practical deployment.

This rapid update ability addresses challenges posed by changing channel conditions or scenarios where initially collected training data is insufficient.

Moreover, this feature is vital in real-world applications where gathering extensive new training datasets is often difficult or time-consuming.

Visual results in Figure 4 confirm that even with limited training samples, DS²C² maintains robust performance and image reconstruction quality, highlighting its efficiency and adaptability in dynamic environments.

Conclusion and Future Work

- | | | |
|----|------------------------------------|---|
| 01 | Robust Semantic Transmission | DS ² C ² provides a deep learning-based multi-layer image transmission scheme at the application layer, resilient to end-to-end channel errors with only moderate standard modifications required. |
| 02 | Disjoint Design Adaptability | The modular disjoint architecture combining RGAN, BResNet, and SumNet enables flexible training, easy updates, and adaptability across varying channel error scenarios. |
| 03 | Future Work Directions | Potential expansions include extending DS ² C ² to video transmission, exploring diverse and realistic channel models, and optimizing the system for specific hardware constraints to enhance deployment feasibility. |
| 04 | Significance for Next-Gen Networks | DS ² C ² 's approach aligns with 6G goals by enabling semantic communication and efficient data transmission, marking a step toward more intelligent and robust multimedia communications. |

Revolutionizing Communication with Semantic Communications



Project Goal: Beyond Bit-Level to Meaning-Centric Communication

Traditional communication systems excel at reliably transmitting bits. However, in an increasingly intelligent and data-rich world, mere bit-fidelity is often insufficient. **The core goal of this project is to explore and develop "Semantic Communications" (SemComs)** – a paradigm shift where communication systems prioritize the *meaning* and *relevance* of information, rather than just raw bits. This approach promises ultra-efficient, highly robust, and task-specific communication, especially vital for bandwidth-constrained, latency-critical, or intelligent applications (e.g., autonomous systems, extended reality, IoT).

Project Contents: Key Research Avenues

- Diverse Data Modalities	- Multi-Disciplinary Research
- Realistic Channel & Network Effects	- Essential Skills & Tools
- Innovative Coding Methodologies	

Starting Point for Quick Engagement

For a fast start, students can begin by addressing the open research questions and expanding upon the foundational work presented in recent key publications, such as the referenced paper [Yetis]. This offers a clear path to immediate contribution and understanding of the current state-of-the-art.

Contact Persons

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Thank you
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Q&A